Sakshyam Sigdel

sakshyam.sigdel@usm.edu | LinkedIn | GitHub | Website | Hattiesburg, MS

Education

University of Southern Mississippi — Expected May 2028
B.S. in Computer Engineering | GPA: 3.79 — *Hattiesburg, MS*

Experience

MLH Production Engineering Fellow (Meta) — June 2026 – Sep. 2026
Major League Hacking × Meta — *Remote*

- Built an open-source portfolio site template (Python, Flask, Jinja, MySQL, Nginx) with unittest suites and CI/CD-automated testing and deployment.
- Administered Linux hosts (systemd, permissions, networking) and debugged live performance issues with vmstat, iostat, strace, lsof, ss, tcpdump, and journalctl.
- Debugged production-style incidents in a 13-fellow pod mentored by Meta engineers; presented the pod's /proc chapter.

Undergraduate Researcher — Computer Vision & Edge AI — Jan. 2026 – Present
AIMS Lab @ USM — AI & Multimodal Systems, Dr. Rabab Abdelfattah — *Hattiesburg, MS*

- Built a two-tier wildfire cascade: a 1 ms CNN smoke gate plus a QDoRA-tuned Gemma 4 VLM emitting structured JSON, trained without manual labels; runs on a Jetson Orin Nano at 15.6 W.
- Found 49% of D-Fire's test set leaks into training; on 9,201 boreal drone frames, VLM verification of CNN alerts cut false alarms **0.639 → 0.031 at 0.990 recall**.

President, Open Source Club — Aug. 2026 – Present
University of Southern Mississippi — *Hattiesburg, MS*

- Grew membership across majors through tabling and intro-to-open-source sessions.
- Run hands-on workshops on Linux, operating systems, and software freedom.

Instructor in Computer Science — June 2026 – Aug. 2026
iD Tech — *Remote*

- Taught K–12 students computer science and engineering 1:1 and in small groups (Python, C++, JavaScript, Arduino, ML/AI), adapting pace from first-time coders to AP level.

Open Source & Projects

procps-ng — Linux `ps`/`top`/`vmstat` | *C* — merged: MR !310

- Fixed vmstat reporting context switches ∼800× too low on its first line, a bug in every release since 2023; one-word patch **merged upstream**.

vimhjkl — Open-Source Vim Trainer | *Python, Vim/Neovim* — github.com/S-Sigdel/vimhjkl

- Built a TUI app that drills advanced vim keystrokes in real vim/nvim — 66 skills, 230 hand-vetted challenges, **540+ GitHub stars**.
- Scheduled reviews with Leitner spaced repetition and belt-rank progression; shipped zero-dependency (pure stdlib) via Homebrew and AUR with CI-automated releases.

Curtain — URL Shortener & Analytics | *Flask, Redis, PostgreSQL, Docker* — github.com/S-Sigdel/Curtain

- **Won Meta PE Hackathon** Scalability Track (team of 2): load-balanced Flask tier via Nginx across Gunicorn containers; split Redis into caching, ID-allocation, and click-tracking roles.
- Validated **500 concurrent users at 0.00% errors, 1.01s p95** with k6; wired Prometheus, Grafana, and Discord alerting with a resilient PostgreSQL fallback.

SenseNav — Assistive Navigation Device | *Arduino, Ultrasound, MiDaS* — github.com/S-Sigdel/SenseNav

- **Top 12 of 320+ teams** at HackMIT: fused ultrasonic ranging with MiDaS monocular depth to render surroundings as spatial audio and haptics for visually impaired users.

Technical Skills

Languages: Python, C/C++, Rust, JavaScript, Bash, SQL
Systems & Infra: Linux, systemd, Docker, Nginx, GitHub Actions, Prometheus, Grafana, k6, GDB, strace, tcpdump
Data & ML: PostgreSQL, MySQL, Redis, Flask, FastAPI, Ultralytics YOLO, TensorRT, llama.cpp, QDoRA
Awards: Midnight Hackathon AI Track; HackPrinceton S25 Best Blockchain + AI; Hack@Davidson 2nd Overall